

1.3.11 Multiplier

Originally, all the components on a system operated at the same frequency. But with improvements in technology, the rest of the system became a bottleneck to the CPU achieving its full potential. This problem was solved by allowing the CPU to operate multiple cycles for every cycle that the system operated. This factor is called the Multiplier. Earlier it was possible for the user to set the multiplier using switches on the motherboard, but now, except for the high-end CPUs, most CPUs come with a set multiplier.

1.3.12 Package types

Every CPU is packaged to ensure that its core components are protected, while allowing connection with the motherboard. Package types seen in current CPUs include PGA, BGA, and LGA.

1.3.13 PGA

Short for Pin Grid Array, this type of packaging has a grid of conducting pins projecting from the package, which connect into the socket. PGA is available in different formats like FC PGA for Flipped Chip PGA, mPGA for micro PGA, and OPGA or organic PGA. AMD CPUs use this form of package.

1.3.14 Pipelining

This refers to the internal data processing pathways of a CPU. The CPU processes data through different stages, like Fetch, Decode, Execute, and Store. By Pipelining, it becomes possible to perform these stages independently and on different bits of data. This allows for more efficient use of CPU cycles.

Modern CPUs have increased the number of stages in the pipeline, and also use advanced techniques like branch prediction, which is used to estimate the possible outcome of an if/else condition, to ensure that every stage of the pipeline is utilised.

The performance of the CPU is greatly influenced by the way it implements pipelining.